McMurdo Station, Ross Island

BUILDING INFORMATION SHEET

CSEC Emergency GENSE

Location ID: B002

Houses a 400kW Caterpillar diesel-engine generator connected to the CSEC (Building 1) to provide emergency back-up power. Building 2 also supplies power to Building 1's emergency lighting, air handling units, heat exchangers, boilers, walk-in freezers, and environmental rooms. It is a single-room building with one personnel door and one overhead roll-up door. A 2-ton trolley crane is provided on an overhead monorail.



Photo 1: *CSEC Emergency GENSE*

General Structure Information

Usage	CSEC Standby Generator Building
Occupancy Group	B-4 Power Plant
Maximum Occupancy	2
Construction Type	Type II, 1-hour, non-combustible; steel-framed structure with 2-1/2" thick, foam-insulated, sandwich panel siding and roofing. Precast concrete floor over steel framing, no interior partitions.
Year Constructed	1993
Dimensions	20' x 25'

Floor Area

Floor	Area
Floor 1	500 sq ft
Total	500 sq ft

HVAC System

Heating	Two 27 kW electric duct heaters in combustion air intake, two Trane 3.3 kW electric unit heaters

Air Handling Units	One thermostatically controlled cooling fan
Fuel Tank	None

Water and Sewer

Water Service	Not connected
Sewer Service	Not connected
Water Heater	None

Electrical Service

Service Feeder	Feeder E
Electrical Service	208/120V, 3PH, fed from 200A breaker in CSEC (Building 1) emergency main switchboard

Fire System

Fire Panel Type	Ansul Auto Pulse 2000 control panel interfaced with one zone on Building 1's King Fisher KFRTI-WP-8 radio interface/transmitter panel,

	pull station, alarm horns
Heat Detectors	Located throughout building
Suppression Type	Ansul total-flood CO2 automatic fire suppression system with four main and four reserve cylinders (75 lb). Ansul MAX back-pressure actuation valves and solenoid release are controlled by a signal from Ansul control panel. The heat detector causes alarms, time delay, and then the release of CO2.

Floorplans

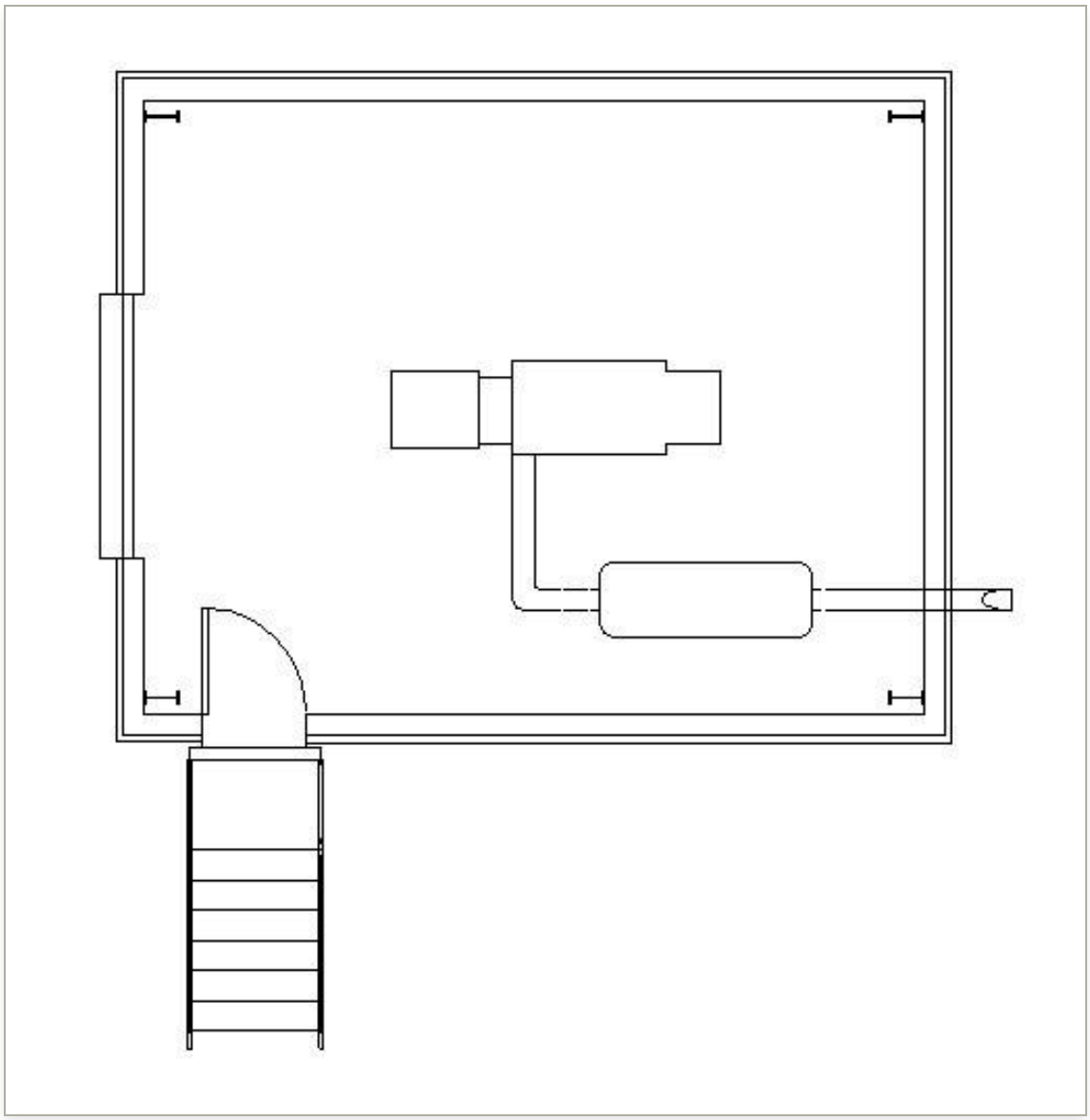


Figure 1: *CSEC Emergency GENSE*